Projeto 1 - Teoria Eletrônica Digital

RA = 2520037944 ultimos 8 algarismos 52003794 (1 2 3 4 5 6 7 8)

$W_a = 5+3 = 8+1 = 9$

$W_B = 2+7 = 9+1 = 10$

$W_c = 0+9 = 9+1 = 10$

$W_d = 0+9 = 4+1 = 5$

$W_T = 9+10+10+5 = 34$

$T_a = \left\lceil \frac{W_T}{3} \right| = 11.33 = 12$ $\quad R = A9 + B10 + C10 + D5$

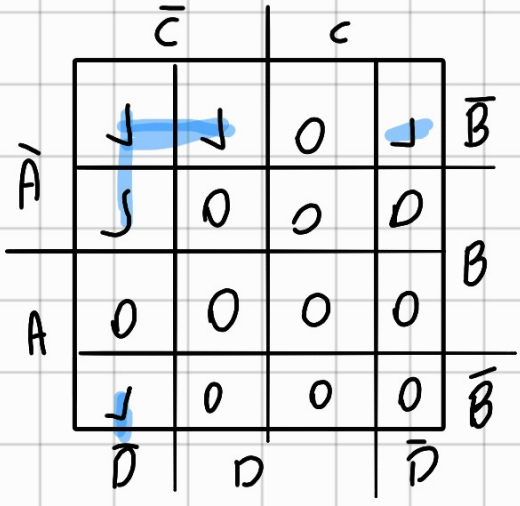
$T_c \left| \frac{2W_T}{3} \right| = 22.66 = 23$

Tabela verdade

A	B	C	D	normal	Alerta	Crítico	R
0	0	0	0	1	0	0	0
0	0	0	1	1	0	0	5
0	0	1	0	1	0	0	10
0	0	1	1	0	1	0	15
0	1	0	0	1	0	0	10
0	1	0	1	0	1	0	15
0	1	1	0	0	1	0	20
0	1	1	1	0	0	1	25
1	0	0	0	1	0	0	9
1	0	0	1	0	1	0	14
1	0	1	0	0	1	0	19
1	0	1	1	0	0	1	24
1	1	0	0	0	1	0	19
1	1	0	1	0	0	1	24
1	1	1	0	0	0	1	29
1	1	1	1	0	0	1	34

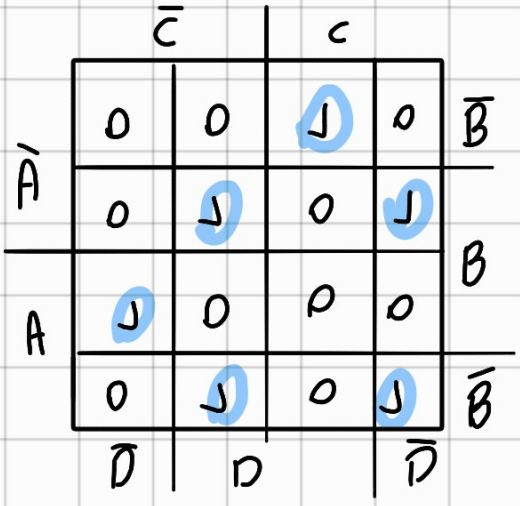
Mapa de Karnaugh

Normal

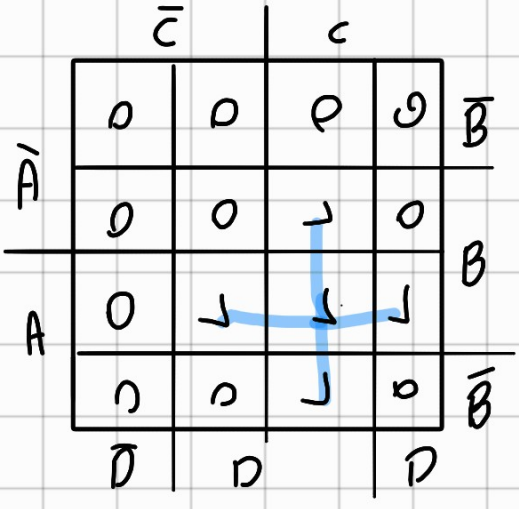


4 duplas: $\bar{A}\bar{C}\bar{B} + \bar{A}\bar{C}\bar{D} + \bar{B}\bar{C}\bar{D} + \bar{B}\bar{D}\bar{A}$

Alta



6 → $AB\bar{C}\bar{D} + \bar{A}B\bar{C}D + A\bar{B}\bar{C}D + \bar{A}\bar{B}CD + \bar{A}BC\bar{D} + A\bar{B}C\bar{D}$



4 duplas: $ABD + BCD + ACD + ABC$